

1. (6 pts) Find the value of b such that $\begin{vmatrix} 3 & -3 & 0 & 2 \\ 1 & 0 & 0 & 3 \\ 0 & 4 & b & 2 \\ -3 & 0 & 0 & 1 \end{vmatrix} = 10$

$$10 = b \begin{vmatrix} 3 & -3 & 2 \\ 1 & 0 & 3 \\ -3 & 0 & 1 \end{vmatrix} = b(3) \begin{vmatrix} 1 & 3 \\ -3 & 1 \end{vmatrix} = 3b(1+9) = 30b \Rightarrow \boxed{b = \frac{1}{3}}$$

2. (8 pts) Given that $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = 4$ find the determinant of each of the following matrices.

a) $\begin{vmatrix} g & h & i \\ d & e & f \\ a & b & c \end{vmatrix} = -4$ b) $\begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix} = 4$ c) $\begin{vmatrix} 2a & 2b & 2c \\ -d & -e & -f \\ \frac{g}{5} & \frac{h}{5} & \frac{i}{5} \end{vmatrix} = -\frac{8}{5}$ d) $\begin{vmatrix} a & b & c \\ 2a & 2b & 2c \\ g & h & i \end{vmatrix} = 0$

3. (10 pts) A is a 9×12 matrix with rank 7. Give the dimension of these subspaces.

a) $\text{Col}(A)$ b) $\text{Col}(A^T)$ c) $(\text{Row}(A))^\perp$ d) $\text{Null}(A^T)$ e) $\text{Null}(A)$

7 7 5 2 5

4. (6 pts) If A is an invertible matrix with eigenvalue λ and corresponding eigenvector $\vec{x}$ prove that $\frac{1}{\lambda}$ is an eigenvalue for A^{-1} .

Given $A\vec{x} = \lambda\vec{x}$

$$A^{-1}(A\vec{x}) = A^{-1}(\lambda\vec{x})$$

$$(A^{-1}A)\vec{x} = \lambda \cdot A^{-1}\vec{x}$$

$$\frac{1}{\lambda} \cdot I\vec{x} = A^{-1}\vec{x}$$

$$\therefore A^{-1}\vec{x} = \frac{1}{\lambda}\vec{x}$$

$$\therefore \frac{1}{\lambda} \text{ is an e-value for } A^{-1} \text{ (with same e-vector } \vec{x})$$

5. (10 pts) Consider the matrix $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 5 & -10 \\ 1 & 0 & 2 & 0 \\ 1 & 0 & 0 & 3 \end{bmatrix}$ with eigenvalues 1, 2, and 3.

a) (6 pts) Find a basis for the eigenspace corresponding to $\lambda_1=1$.

e-space = Nullspace for $A - \lambda_1 I$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & -10 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 2 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = -2x_4$$

$$x_2 = x_2$$

$$x_3 = 2x_4$$

$$x_4 = x_4$$

$$\text{Basis} = \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 2 \\ 1 \end{bmatrix} \right\}$$

b) (4 pts) Is the matrix A diagonalizable or not? Give a reason for your answer.

Yes: geometric multiplicity = algebraic mult. for all e-values.

$$\text{For } \lambda_1 \text{ geo. mult.} = 2$$

$$\lambda_2 \text{ geo mult} = 1$$

$$\lambda_3 \text{ geo mult.} = 1$$

6. (12 pts) Let $\mathbf{w}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$ and $\mathbf{w}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$ and let $W = \text{span}(\mathbf{w}_1, \mathbf{w}_2)$.

a) Find a vector $\mathbf{v}$ that is in W and is orthogonal to $\mathbf{w}_1$.

$$\begin{aligned} \mathbf{v} &= \mathbf{w}_2 - \text{Proj}_{\mathbf{w}_1} \mathbf{w}_2 = \mathbf{w}_2 - \frac{\mathbf{w}_2 \cdot \mathbf{w}_1}{\mathbf{w}_1 \cdot \mathbf{w}_1} \mathbf{w}_1 \in \text{span}(\mathbf{w}_1, \mathbf{w}_2) \\ &= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} -\frac{1}{3} \\ -\frac{1}{3} \\ \frac{2}{3} \\ 1 \end{bmatrix} \end{aligned}$$

b) Find a basis for the orthogonal complement $W^\perp$ Let $A = [\vec{w}_1, \vec{w}_2] = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$

$$W = \text{span}(\mathbf{w}_1, \mathbf{w}_2) = \text{Col}(A) = \text{Row}(A^T)$$

$$W^\perp = \text{Null}(A^T)$$

Find a basis for $\text{Null}(A^T)$

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

$$x_1 = -x_2 + x_4$$

$$x_2 = x_2$$

$$x_3 = -x_4$$

$$x_4 = x_4$$

$$\text{Basis for } W^\perp \text{ is } \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix} \right\}$$

7. (8 pts) Show that the set of vectors $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \right\}$ form an orthogonal basis for

$\mathbb{R}^3$ and find the coordinate vector of $\begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix}$ with respect to that basis.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & -1 \\ 1 & -1 & 1 & 2 \\ 1 & 0 & -2 & 5 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3/2 \\ 0 & 0 & 1 & -3/2 \end{array} \right]$$

$$\begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 \\ -3/2 \\ -3/2 \end{bmatrix}_{\mathcal{B}}$$

vectors in $\mathcal{B}$ are a basis for $\mathbb{R}^3$ since they are 3 linearly independent vectors in $\mathbb{R}^3$.

Orthogonal since $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = 0$, $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} = 0$ and $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} = 0$

[Could also find coordinates using projections]

8. (10 pts) Let $\mathbf{v} = \begin{bmatrix} -1 \\ 10 \\ 3 \end{bmatrix}$ and $W = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$. Find the orthogonal decomposition of $\mathbf{v}$

with respect to W . $\mathbf{v} = \mathbf{w} + \mathbf{w}^\perp$ Let $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$$\mathbf{w} = \text{proj}_{\mathbf{u}} \mathbf{v} = \frac{\mathbf{v} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u} = \frac{12}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix}$$

$$\mathbf{w}^\perp = \mathbf{v} - \mathbf{w} = \begin{bmatrix} -1 \\ 10 \\ 3 \end{bmatrix} - \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} -5 \\ 6 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} -1 \\ 10 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix} + \begin{bmatrix} -5 \\ 6 \\ -1 \end{bmatrix}$$

9. (10 pts) Recall that $\mathcal{P}_2$ is the set of all polynomials with real coefficients and degree at most 2. Let $W = \{a + bx + cx^2 : a+b+c = 0\}$. Prove that W is a subspace of $\mathcal{P}_2$.

① W consists of polynomials of degree ≤ 2 so is a subset of $\mathcal{P}_2$
 $1+x-2x^2 \in W$ since $1+1-2=0$. $\therefore W$ is not empty

② Let $f(x) = a_1 + b_1x + c_1x^2$ and $g(x) = a_2 + b_2x + c_2x^2$ be in W
 so $a_1 + b_1 + c_1 = 0$ and $a_2 + b_2 + c_2 = 0$.

$\therefore f(x) + g(x) = (a_1 + a_2) + (b_1 + b_2)x + (c_1 + c_2)x^2 \in W$ because

$$a_1 + a_2 + b_1 + b_2 + c_1 + c_2 = (a_1 + b_1 + c_1) + (a_2 + b_2 + c_2) = 0 + 0 = 0.$$

$\therefore W$ is closed under addition.

③ Let k be any scalar.

$k \cdot f(x) = ka_1 + kb_1x + kc_1x^2 \in W$ because

$$ka_1 + kb_1 + kc_1 = k(a_1 + b_1 + c_1) = k \cdot 0 = 0.$$

$\therefore W$ is closed under scalar mult.

10. (8pts) The set of polynomials $\{1+x, x+x^2, 1+x^2\}$ is a basis for $\mathcal{P}_2$. Find the coordinate vector for $1+2x+3x^2$ with respect to this basis.

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 3 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{array} \right]$$

The coordinate vector for $1+2x+3x^2$ with respect to $\mathcal{B}$

$$\text{is } \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}_{\mathcal{B}}$$

check

$$1+2x+3x^2 = 0(1+x) + 2(x+x^2) + 1(1+x^2)$$

11. (6 pts) Suppose A and B are $n \times n$ orthogonal matrices. Prove that the product AB is orthogonal.

Given A and B are orthogonal means $A^{-1} = A^T$ and $B^{-1} = B^T$

$$\therefore (AB)^{-1} = B^{-1}A^{-1} = B^T A^T = (AB)^T$$

$\therefore AB$ is orthogonal

12. (6 pts) What is the dimension for each of the following vector spaces?

a) $\mathcal{P}_n$

b) $\mathcal{M}_{jk}$

c) The set of upper triangular 3×3 matrices

(a) $\dim(\mathcal{P}_n) = n+1$ (b) $\dim(\mathcal{M}_{jk}) = j \cdot k$ (c) $\dim = 6$

$$\begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$$